Ex.No:07 Sliding Window Protocol

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Aim:

To write sliding window protocol code using python

Code:

Sender.py

```
import time
import os
ws = input("window size")
msg = input("Enter message: ")
frame = [(i+1, ch) for i, ch in enumerate(msg)]
i = 0
while i < len(frames):
    window = frames[i:i+ws]
    with open("Sender_Buffer", "w") as f:
        for no, data in window:
            f.write(f"{no}, {data}\n")
    print("\n Sending :", window)
    time.sleep(2)
    while not os.path.exists("Reciver_Buffer"):
        time.sleep(1)
    with open("Receiver_Buffer") as f:
        ack, no = f.read().strip().split();
        no = int(no)
    print("Received", ack, no)
    if ack == "ACK" and no == window[-1][0]:
        i = ws
    elif ack == "NACK":
        i = no - 1
    open("Receiver_Buffer", "w").close()
print("\n Transmission completed")
```

Receiver.py

```
import time,os
expected = 1
while True:
    if not os.path.exists("Sender_Buffer"):
        time.sleep(1)
        continue
    with open("Sender_Buffer") as f:
        frames = [int(x.split(",")[0])
                  x.split(",")[1].strip() for x
                  in f if x.strip()]
    if not frames:
        time.sleep(1)
        continue
    print("\n Received ", frames)
    for (no, data) in frames:
        if no != expected:
            with open("Receiver_Buffer",
                      "w") as f:
                f.write(f"NACK,{expected}")
            break
        expected += 1
    else:
        with open("Receiver_Buffer",
                  "w") as f:
            f.write(f"ACK,{frames[-1]}")
    time.sleep(1)
```

Result:

Thus, the sliding window protocol was implemented successfully.